

3 WORM COLORINGS

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14 Abstract

15 Given a coloring of the vertices, we say subgraph H is monochromatic
16 if every vertex of H is assigned the same color, and rainbow if no pair of
17 vertices of H are assigned the same color. Given a graph G and a graph F ,
18 we define an F -WORM coloring of G as a coloring of the vertices of G
19 without a rainbow or monochromatic subgraph H isomorphic to F . We
20 present some results on this concept especially as regards to the existence,
21 complexity, and optimization within certain graph classes. The focus is on
22 the case that F is the path on three vertices.

23 **Keywords:** Coloring, rainbow, monochromatic, forbidden, path.

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25 1. INTRODUCTION

26 Let F be a graph. Consider a coloring of the vertices of G . We say that a copy
27 of F (as a subgraph) is *rainbow* if all its vertices receive different colors. We say
28 that a copy of F (as a subgraph) is *monochromatic* if all its vertices receive the

same color. It is easy to avoid monochromatic copies: color every vertex in G a different color. It is also easy to avoid rainbow copies: color every vertex the same color. But things are more challenging if one tries to avoid both simultaneously. For example, if $G = K_5$, then any coloring of G yields either a monochromatic or a rainbow P_3 . On the other hand, if we color $G = K_4$ giving two vertices red and two vertices blue, we avoid both a rainbow and a monochromatic P_3 .

So we define an *F-WORM coloring* of G as a coloring of the vertices of G WithOut a Rainbow or Monochromatic subgraph H isomorphic to F . We assume the graph F has at least 3 vertices, since any subgraph on 1 or 2 vertices is automatically rainbow or monochromatic. For example, if G is bipartite (and F is not empty), then the bipartition is automatically an *F-WORM* coloring. Indeed, if G is k -colorable with k less than the order of F (and F is nonempty), then a proper k -coloring of G is an *F-WORM* coloring.

In this paper we explore the concept and establish some basic properties. We also consider, for a graph that has such a coloring, what the range of colors is. To this end, we define $W^+(G, F)$ as the maximum number of colors and $W^-(G, F)$ as the minimum number of colors in an *F-WORM* coloring of graph G . In this paper we focus on the fundamental results and the case that F is the path P_3 . Some further results where a cycle or clique is forbidden are given in [6].

Vertex colorings with various local constraints, especially avoiding monochromatic subgraphs, have been studied extensively; see for example [10]. Edge-colorings that avoid rainbow subgraphs have been studied under the term “anti-Ramsey numbers”; see for example [2]. There are also a few papers on edge-colorings that avoid some monochromatic and some rainbow subgraph; see for example [1].

More recently, vertex colorings that avoid rainbow subgraphs have been considered by Bujtás et al. [4], whose *3-consecutive C-coloring* is equivalent to a coloring without a rainbow P_3 , and by Bujtás et al. [3] who defined the *star-[k] upper chromatic number* as the maximum number of colors in a coloring of the vertices without a rainbow $K_{1,k}$. It follows that $W^+(G, P_3)$ is at most the star-[2] upper chromatic number. However, our parameter is not equal to theirs (even in graphs where $W^+(G, P_3)$ exists). For example, take the tree S obtained from the star on three edges by subdividing each edge once. Then one can color S with 4 colors while avoiding a rainbow P_3 (color the center and all its neighbors the same color, and color each leaf with a different color). On the other hand, one can easily check that any P_3 -WORM coloring of S uses at most three colors (or see Theorem 17 below).

We proceed as follows. In Section 2 we show that if a graph has a P_3 -WORM coloring then it has one using two colors, from which it follows that the decision problem is NP-hard. In Sections 3 and 4 we consider the existence and range of P_3 -WORM colorings for several graph families including bipartite

70 graphs, Cartesian products, cubic graphs, outerplanar graphs, and trees. Finally,
 71 we consider some related complexity results in Section 5 and an extremal question
 72 in Section 6.

73 2. BASICS

74 In this section we consider the maximum and minimum number of colors in a
 75 P_3 -WORM coloring. In particular, we show that if a graph has a P_3 -WORM
 76 coloring then it has such a coloring with only two colors. Note that we consider
 77 only connected graphs G , since if G is disconnected then the existence and range of
 78 colors for G is determined by the existence and range of colors for the components.
 79 For example, $W^+(G, P_3)$ is the sum of $W^+(G_i, P_3)$ over all components G_i of G .

80 We consider first the maximum number of colors that a P_3 -WORM coloring
 81 may use.

82 **Theorem 1.** *If a graph G on n vertices has a P_3 -WORM coloring, then*
 83 $W^+(G, P_3) \leq n/2 + 1$.

84 **Proof.** Note that if we add edges to the graph, then the constraints increase,
 85 and so W^+ can only decrease. So it suffices to prove the result for G a tree. The
 86 result is by induction. The base case of $n = 1$ is trivial. Further, if G is a star
 87 then $W^+(G, P_3) = 2$; so we may assume that G is not a star. Let v be a non-leaf
 88 vertex that has at most one non-leaf neighbor w . Let $G' = G - \{v\} - L_v$, where
 89 L_v is the set of leaf-neighbors of v . Then G' is connected. Further, $N_G[v]$ receives
 90 at most two colors, and if exactly two colors, then one of those colors is the same
 91 color as w . It follows that the number of colors in G is at most one more than
 92 the number of colors in G' , and the bound follows. ■

93 One example of equality in Theorem 1 is the case that G is a path:

94 **Observation 2.** *For the path on n vertices, $W^+(P_n, P_3) = \lfloor n/2 \rfloor + 1$.*

95 **Proof.** Say the vertices of the path are $v_1 v_2 \dots v_n$. Color v_1 with color 1, color
 96 v_2 and v_3 with color 2, color v_4 and v_5 with color 3, and so on. This coloring has
 97 neither a rainbow nor a monochromatic P_3 and uses $\lfloor n/2 \rfloor + 1$ colors. ■

98 The above theorem can also be deduced from a result of Bujtas et al. [4].
 99 They showed that their 3-consecutive C -coloring number of a connected graph is
 100 at most one more than the vertex cover number, which we denote by $\beta(G)$. By
 101 definition, a P_3 -WORM coloring is a 3-consecutive C -coloring. Thus:

102 **Observation 3.** *If a connected graph G has a P_3 -WORM coloring, then*
 103 $W^+(G, P_3) \leq \beta(G) + 1$.

104 We next consider the minimum number of colors that a P_3 -WORM coloring
105 may use.

106 **Theorem 4.** *A graph G has a P_3 -WORM coloring if and only if G has a P_3 -*
107 *WORM coloring using only two colors.*

108 **Proof.** Consider a P_3 -WORM coloring of the graph G . Say an edge is monochro-
109 matic if its two ends have the same color. By the lack of monochromatic P_3 's,
110 the monochromatic edges form a matching. Let H be the spanning subgraph of
111 G with the monochromatic edges removed. Consider any edge uv in H ; say u
112 is color i and v is color j . Then by the lack of rainbow P_3 's, every neighbor of
113 u is color j and every neighbor of v is color i . It follows that H is bipartite.
114 If we 2-color G by the bipartition of H , the monochromatic edges still form a
115 matching, and so this is a P_3 -WORM coloring. ■

116 Recall that a 1-defective 2-coloring of a graph G is a 2-coloring such that
117 each vertex has at most one neighbor of its color. It follows that:

118 *A 1-defective 2-coloring is equivalent to a P_3 -WORM 2-coloring.*

119 For example, Cowen [5] proved that determining whether a graph has a 1-defective
120 2-coloring is NP-complete. It follows that determining whether a graph has a P_3 -
121 WORM coloring is NP-complete.

122 It is not true, however, that if a graph has a P_3 -WORM coloring using
123 k colors, then it has one using j colors for every $2 < j < k$. Indeed, we now
124 construct a graph H_k that has a P_3 -WORM coloring using k colors and one using
125 2 colors, but for no other number of colors.

126 For $k \geq 3$ we construct graph H_k as follows. Let $s = \max(3, k-2)$. For every
127 ordered pair of distinct i and j , with $i, j \in \{1, \dots, k\}$, create disjoint sets B_i^j of s
128 vertices. For each i define $C_i = \bigcup_{j \neq i} B_i^j$. Then add all s^2 possible edges between
129 sets B_i^j and B_j^i for all $i \neq j$. For each triple of distinct integers i, j, j' , add exactly
130 one edge between B_i^j and $B_i^{j'}$ such that for each i the subgraph induced by C_i
131 has maximum degree 1. One possibility for the graph H_4 is shown in Figure 1.

132 **Observation 5.** *For $k \geq 3$, every P_3 -WORM coloring of H_k uses either 2 or k*
133 *colors.*

134 **Proof.** Consider a P_3 -WORM coloring of H_k . Note that the subgraph induced
135 by $B_i^j \cup B_j^i$ is $K_{s,s}$. It is easy to show that for $s \geq 3$ the only P_3 -WORM coloring
136 of $K_{s,s}$ is the bipartition. It follows that for each i and j , all s vertices in B_i^j
137 receive the same color; further, the color of B_i^j is different from the color of B_j^i .
138 Because there is a P_3 that goes from B_j^i to B_i^j to $B_i^{j'}$, it must be that $B_i^{j'}$ receives

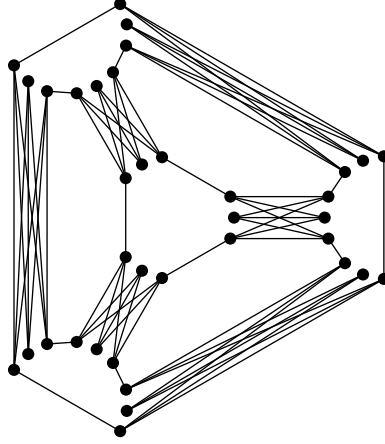


Figure 1. The graph H_4 whose P_3 -WORM colorings use either 2 or 4 colors

139 either the color of B_i^j or the color of B_j^i . That is, there are precisely two colors
 140 on all of $C_i \cup C_j$.

141 So suppose some C_i receives two colors. Then every other C_j is colored with
 142 a subset of these two colors. Otherwise, assume every C_i is monochromatic. It
 143 follows that every C_i is a different color, and thus we use k colors. ■

144 3. SOME CALCULATIONS

145 Now we consider P_3 -WORM colorings for some specific families of graphs.

146 3.1. Bipartite graphs

147 As observed earlier, the bipartite coloring of a bipartite graph is automat-
 148 ically a P_3 -WORM coloring. So we focus on the maximum number of colors a
 149 WORM-coloring may use. We observed above that every P_3 -WORM coloring of
 150 $K_{m,m}$ uses two colors. Indeed, we now observe the following slightly more general
 151 result:

152 **Observation 6.** For $n \geq m \geq 2$, $W^+(K_{n,n}, K_{1,m}) = 2m - 2$.

153 **Proof.** One can achieve $2m - 2$ colors by using disjoint sets of $m - 1$ different
 154 colors on each partite set. So we need to prove the upper bound.

155 Let A and B denote the partite sets of $K_{n,n}$ and consider a $K_{1,m}$ -WORM
 156 coloring. Suppose that one partite set receives at least m different colors, say A .
 157 Then starting with these m vertices, it follows that every vertex v in B must be
 158 one of these m colors. Furthermore, v cannot see m distinct colors different from

159 it. That is, the coloring uses exactly m colors. On the other hand, if every partite
 160 set has at most $m - 1$ colors, the total number of colors is at most $2(m - 1)$. It
 161 follows that $W^+(K_{n,n}, K_{1,m}) \leq \max(m, 2m - 2) = 2m - 2$. ■

162 We saw earlier that $W^+(P_n, P_3) = \lfloor n/2 \rfloor + 1$. This result can be generalized
 163 slightly to other forbidden paths:

164 **Theorem 7.** *For $n \geq m \geq 3$, $W^+(P_n, P_m) = \left\lfloor \frac{(m-2)n}{m-1} \right\rfloor + 1$.*

165 **Proof.** Let the path P_n be $v_1 v_2 \dots v_n$. Give every vertex a different color except
 166 that $v_{a(m-1)}$ and $v_{a(m-1)+1}$ receive the same color for $1 \leq a \leq \lfloor (n-1)/(m-1) \rfloor$.
 167 For example, if $m = 4$, then the coloring of P_n starts 1, 2, 3, 3, 4, 5, 5, 6, 7, 7, 8, 9, ...
 168 Thus, the total number of colors is $n - \lfloor (n-1)/(m-1) \rfloor$, which equals the
 169 claimed formula. It is easily checked that the coloring has neither a rainbow nor
 170 a monochromatic P_m .

We next prove the upper bound by induction on n for fixed m . The base
 cases are $n \leq 2m - 2$. For these n , $\lfloor (m-2)n/(m-1) \rfloor + 1 = n - 1$, and the
 desired conclusion is true. Now let $n \geq 2m - 1$. By the induction hypothesis, the
 number of colors used by the first $n - m + 1$ vertices of P_n is at most

$$\left\lfloor \frac{(m-2)(n-m+1)}{(m-1)} \right\rfloor + 1 = \left\lfloor \frac{(m-2)n}{(m-1)} \right\rfloor - m + 3.$$

171 Also note that the last $m - 1$ vertices of P_n use at most $m - 2$ colors other than
 172 those used by the first $n - m + 1$ vertices, otherwise we would have a rainbow P_m .
 173 Therefore, the total number of colors used is at most $\lfloor (m-2)n/(m-1) \rfloor + 1$.
 174 This completes the proof. ■

175 3.2. Cartesian products

176 Recall that the *Cartesian product* of graphs G and H , denoted $G \square H$, is the
 177 graph whose vertex set is $V(G) \times V(H)$, in which two vertices (u_1, u_2) and (v_1, v_2)
 178 are adjacent if $u_1 v_1 \in E(G)$ and $u_2 = v_2$, or $u_1 = v_1$ and $u_2 v_2 \in E(H)$. We next
 179 consider a P_3 -WORM coloring of $G \square H$.

180 **Theorem 8.** *If G and H are nontrivial connected graphs and $G \square H$ has a P_3 -*
 181 *WORM coloring, then it uses only two colors.*

182 **Proof.** It suffices to prove the result when G and H are trees. We proceed by
 183 induction. Clearly when $G = H = K_2$, we have $W^+(C_4, P_3) = 2$.

184 So assume that at least one of the factors, say G , has order at least 3. Let
 185 u be a leaf of G , with neighbor u' , and let $G' = G - \{u\}$. By the inductive
 186 hypothesis, every P_3 -WORM coloring of $G' \square H$ uses only two colors. Consider
 187 any vertex v of H . Since G is not K_2 , vertex u' has at least one neighbor in G' ,

188 and so vertex (u', v) is the center of a P_3 in $G' \square H$. This means that the vertex
 189 (u', v) has a neighbor x of a different color in $G' \square H$, and thus (u, v) must get
 190 either the color of (u', v) or x . ■

191 3.3. Cubic graphs

192 Let G be a connected cubic graph. We know from [8] that G has a 2-coloring
 193 where every vertex has at most one neighbor of the same color. This coloring is
 194 a P_3 -WORM. So the natural question is: What is the minimum and maximum
 195 value of $W^+(G, P_3)$ as a function of the order n ?

196 There are many cubic graphs G that have $W^+(G, P_3) = 2$. One general
 197 family is the ladder $C_m \square K_2$. (See Theorem 8.)

198 Computer checking of small cases suggests that the maximum value of pa-
 199 rameter $W^+(G, P_3)$ is $n/4 + 1$. This value is achieved by several graphs including
 200 the following graph. For $s \geq 2$ create B_s by taking s copies of $K_4 - e$ and adding
 201 edges to make the graph cubic and connected. For example, B_5 is illustrated in
 Figure 2.

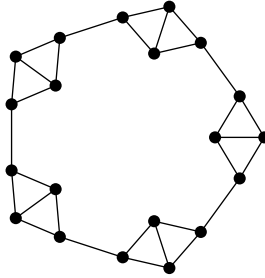


Figure 2. Graph B_5 conjectured to have maximum $W^+(G, P_3)$ for cubic graphs

202

203 **Observation 9.** For $s \geq 2$, $W^+(B_s, P_3) = s + 1$.

204 **Proof.** Consider a P_3 -WORM coloring of B_s . It is easy to show that each copy
 205 of $K_4 - e$ has exactly two colors, and one of those colors is also present at the end
 206 of each edge leading out of the copy. Thus $s + 1$ is an upper bound. An optimal
 207 coloring is obtained by coloring each central pair from a $K_4 - e$ with a new color,
 208 and coloring all other vertices the same color. ■

209 Another interesting case is where the forbidden graph is the star on three
 210 edges. Here the maximum $W^+(G, K_{1,3})$ for cubic graphs G of order n is $3n/4$,
 211 achieved uniquely by the above graph B_s . The upper bound is given by Propo-
 212 sition 18 of [3]:

213 **Theorem 10.** [3] For an r -regular graph G of order n , $W^+(G, K_{1,r}) \leq rn/(r+1)$.

214 3.4. Outerplanar graphs

215 Recall that a *maximal outerplanar graph*, or *MOP*, is an outerplanar graph
 216 with a maximum number of edges. That is, an outer cycle with chords trian-
 217 gulating the interior. In this section, we determine which maximal outerplanar
 218 graphs have a P_3 -WORM coloring. But first we note that if such a graph G has
 219 a P_3 -WORM coloring, then $W^+(G, P_3) = W^-(G, P_3) = 2$.

220 **Observation 11.** *If a MOP has a P_3 -WORM coloring, then that coloring uses*
 221 *two colors.*

222 **Proof.** Consider some triangle $T_0 = \{x, y, z\}$. It must have exactly two colors;
 223 say red and blue. If this is the whole graph we are done. Otherwise, there is
 224 another triangle T_1 that overlaps T_0 in two vertices. Say T_1 has vertices $\{x, y, w\}$.
 225 Then if x and y have different colors, w must be one of their colors. Further, if
 226 x and y are both red say, since zxw is a P_3 it must be that w is the same color
 227 as z . That is, all vertices of T_1 are red or blue. Repeating the argument we see
 228 that all vertices in the graph are red or blue. ■

229 We now consider the necessary conditions for a P_3 -WORM coloring. Note
 230 that by Theorem 4, this is equivalent to determining which MOPs have a 1-
 231 defective 2-coloring. Let F_6 denote the fan given by the join $K_1 \vee P_6$. This graph
 232 and the Hajós graph (also known as a 3-sun) are shown in Figure 3.

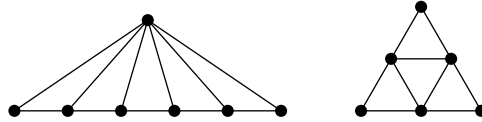


Figure 3. Two MOPs: the fan F_6 and the Hajós graph

233 **Observation 12.** *Neither the fan F_6 nor the Hajós graph has a P_3 -WORM*
 234 *coloring.*

235 **Proof.** Consider a 2-coloring of the fan F_6 . Let v be the central vertex; say v is
 236 colored red. Then at most one other vertex can be colored red. It follows that
 237 there must be 3 consecutive non-red vertices on the path. Thus, the coloring is
 238 not WORM.

239 Consider a 2-coloring of the Hajós graph. It is immediate that two of the
 240 central vertices must be one color, say red, and the other central vertex the other
 241 color, say blue. Now let u and v be the two vertices of degree 2 that have a blue
 242 neighbor. Then, coloring either of them red creates a red P_3 , but coloring both
 243 of them blue creates a blue P_3 . ■

244 So it is necessary that the MOP has maximum degree at most 5 and contains
 245 no copy of the Hajós graph. For example, one such MOP is drawn in Figure 4.
 246 (The vertices of degree 5 are in white.)

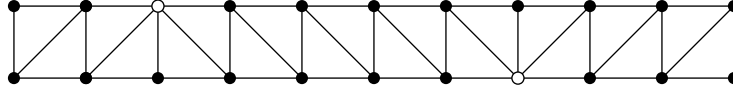


Figure 4. A MOP that has no F_6 or Hajós subgraph

247 The *interior graph* of a MOP G , denoted by C_G , is the subgraph of G induced
 248 by the chords. (This is well-defined as a MOP has a unique Hamiltonian cycle.)

249 **Observation 13.** *A MOP G contains a copy of Hajós graph if and only if the*
 250 *interior graph C_G has a cycle.*

251 **Proof.** If G contains a copy of Hajós graph, then C_G contains a triangle. Con-
 252 versely, assume C_G contains a cycle; then it must contain a triangle. Since every
 253 chord of G is contained in two adjacent triangles, it follows that G contains a
 254 copy of the Hajós graph. ■

255 A *caterpillar* is a tree in which every vertex is within distance one of a central
 256 path. Hedetniemi et al. [7] showed that if the interior graph of a MOP is acyclic,
 257 then it is a caterpillar. Let V_5 denote the set of vertices of degree 5 in a MOP G .
 258 Equivalently, V_5 is the vertices of degree 3 in C_G . Define a *stem* as a path in C_G
 259 whose ends are in V_5 and whose interior vertices are not.

260 **Theorem 14.** *A MOP G has a 1-defective 2-coloring (equivalently a P_3 -WORM*
 261 *coloring) if and only if*

- 262 (a) G has maximum degree at most 5,
- 263 (b) the interior graph C_G is a caterpillar, and
- 264 (c) every stem of C_G has odd length.

265 **Proof.** We first prove necessity. Let G be a MOP with a P_3 -WORM coloring. By
 266 Observations 12 and 13, G has maximum degree at most 5 and C_G is a caterpillar.
 267 Let P denote the central path of the caterpillar C_G . We are done unless C_G has
 268 a stem; so assume $P_{u,v}$ is a stem with ends u and v .

269 Let the path through $N(u)$ be $u_1u_2u_3u_4u_5$. Note that u_1 and u_5 are neighbors
 270 of u on the outer cycle. Further, by the lack of Hajós subgraph, the edge u_2u_3 is
 271 not in C_G ; that is, u_2 and u_3 are consecutive on the outer cycle. Similarly, so are
 272 u_3 and u_4 . It follows that u_3 has degree exactly 3 in G , and so u_3 is a leaf in C_G .

273 Now consider the P_3 -WORM coloring of G . By Observation 11, this coloring
 274 uses two colors, say 1 and 2. It is easy to see that u must have the same color
 275 as u_3 , while u_1, u_2, u_4, u_5 have the other color. In particular, u has no neighbor

on P of the same color. Similarly, v has no neighbor on P of the same color. Further, since the subgraph of G induced by the vertices of $P_{u,v}$ is its square, all other vertices do have neighbors on P of the same color. Indeed, assume u has color 1; then the coloring pattern of $P_{u,v}$ must be either $1, 2, 2, 1, 1, \dots, 2, 2, 1$ or $1, 2, 2, 1, 1, \dots, 1, 1, 2$. Hence, the stem $P_{u,v}$ must have odd length.

Now we prove sufficiency. Assume G has maximum degree at most 5, and the interior graph C_G is a caterpillar with every stem of C_G having odd length. We color P with two colors such that every vertex of V_5 has no neighbor on P of the same color and all other vertices (except possibly the end-vertices of P) do have neighbors on P of the same color. Then we give each vertex of $C_G - P$ the same color as their neighbor in C_G .

It remains to color the (at most two) vertices of degree 2 in G . Let x be a vertex of degree 2 in G , and let y and z be its neighbors. Let t be the other vertex with which y and z forms a triangle. Say t is adjacent to z on the outer cycle. By the construction of the coloring so far, it follows that either y has the same color as t , in which case we can give x the same color as z , or y has the same color as z , in which case we can give x the other color. Thus we can extend the coloring to the whole graph, as required. ■

4. TREES

The following observation will facilitate bounds for $W^+(T, P_3)$ when T is a tree.

Observation 15. *Consider a tree T . A P_3 -WORM coloring of some of the vertices, such that the colored vertices induce a connected subgraph, can be extended to a P_3 -WORM coloring of the whole tree.*

Proof. Assume we have a P_3 -WORM coloring of $U \subseteq V(T)$ such that U induces a connected subgraph of T . Consider any uncolored vertex v that is adjacent to some colored vertex w_v (since T is a tree, w_v is unique). If w_v sees a color c different from its own color, then assign v the color c . If w_v has no colored neighbor or its only colored neighbor has the same color as it, then give v any other color. In both cases we do not create a monochromatic or rainbow P_3 . Repeat until all vertices colored. ■

For example, since a tree T contains a path of the same diameter, it follows from Observation 2 that $W^+(T, P_3) \geq \text{diam}(T)/2 + 1$.

We consider next a tree algorithm. There are general results (see for example [9]) that show that there is a linear-time algorithm to compute the parameter $W^+(T, P_3)$ for a tree T , and indeed for bounded treewidth. Nevertheless, we give the details of an algorithm below, and then use it to calculate the value of

312 $W^+(T, P_3)$ for a spider (sometimes called an octopus). We do the standard pos-
 313 torder traversal algorithm. That is, we root the tree at some vertex r and then
 314 calculate a vector at each vertex representing the values of several parameters on
 315 the subtree rooted at that vertex.

316 For vertex v , define T_v to be the subtree rooted at v and $k(v)$ to be the
 317 number of children of v . Define $p(v)$ to be the maximum number of colors in
 318 a P_3 -WORM coloring of T_v with the constraint that v has a child of the same
 319 color (“partnered”); and define $s(v)$ to be the maximum number of colors in a
 320 P_3 -WORM coloring of T_v with the constraint that v has no child of the same
 321 color (“solitary”). By Observation 15, such a coloring exists (that is, $p(v)$ and
 322 $s(v)$ are defined) except for the case of $p(v)$ when $k(v) = 0$.

323 Define $\ell_p(v) = 2$ if $k(v) \geq 2$ and 1 otherwise; define $\ell_s(v) = 2$ if $k(v) \geq 1$,
 324 and 1 otherwise. Note that $\ell_p(v)$ and $\ell_s(v)$ denote the number of colors in $N[v]$
 325 in a partnered and solitary coloring of T_v respectively. Let $P(v) = p(v) - \ell_p(v)$
 326 and $S(v) = s(v) - \ell_s(v)$.

Theorem 16. *If vertex v has children $c_1, \dots, c_k$, $k \geq 1$, then*

$$p(v) = \begin{cases} \max_{1 \leq i \leq k} \left\{ 1 + s(c_i) + \sum_{j \neq i} \max(P(c_j), S(c_j)) \right\}, & \text{if } k \geq 2, \\ s(c_1), & \text{otherwise;} \end{cases}$$

$$s(v) = 2 + \sum_{i=1}^k \max(P(c_i), S(c_i)).$$

327 **Proof.** Consider a P_3 -WORM coloring of T_v . Say v has children $c_1, \dots, c_k$. To
 328 maximize the colors, the color-set used in T_{c_i} should be as disjoint as possible
 329 from the color-set used in T_{c_j} . But note that there has to be some overlap.

330 Specifically, if v is solitary, then all its children have the same color. In the
 331 tree T_{c_i} , any child of c_i has the same color as either c_i or v . So the maximum
 332 number of colors that appear only in $T_{c_i} - \{c_i\}$ is $\max(P(c_i), S(c_i))$. Further, if
 333 v is partnered, say with c_i , then there are $s(c_i)$ colors in the subtree T_{c_i} . There
 334 is 1 color for all other children c_j of v . As above, the maximum number of colors
 335 that appear only in $T_{c_j} - \{c_j\}$ is $\max(P(c_j), S(c_j))$. ■

336 Since these maxima can be computed in time proportional to $k(v)$, and
 337 $W^+(T, P_3) = \max(p(r), s(r))$, we obtain a linear-time algorithm to calculate
 338 $W^+(T, P_3)$ for a tree T .

339 As an application, we determine the value of $W^+(T, P_3)$ for an octopus:

Theorem 17. *Let X be a star with $k \geq 2$ leaves, and let T be the subdivision
 of X where the i^{th} edge of X is subdivided $a_i \geq 0$ times for $1 \leq i \leq k$. Then*

$$W^+(T, P_3) = 2 + \sum_{i=1}^k \left\lceil \frac{a_i - 1}{2} \right\rceil + x,$$

340 where x is 1 if at least one a_i is odd and 0 otherwise.

341 **Proof.** This follows from Theorem 16 by considering the children $c_1, \dots, c_k$ of
 342 the original center. It is easy to check that $P(c_i) = \lceil (a_i - 1)/2 \rceil$ and $S(c_i) =$
 343 $\lceil (a_i - 2)/2 \rceil$. ■

344 5. WORM IS EASY SOMETIMES

345 We observed earlier that determining whether a graph has a P_3 -WORM coloring
 346 is NP-hard. There are at least a few cases of forbidden graphs where the problem
 347 has a polynomial-time algorithm. The first case is trivial:

348 **Observation 18.** *For $F = mK_1$, graph G has an F -WORM-coloring if and only*
 349 *if G has at most $(m - 1)^2$ vertices.*

350 Here is another forbidden graph with a characterization:

351 **Observation 19.** *Let F be the one-edge graph on three vertices. A graph G has*
 352 *an F -WORM coloring if and only if G is (a) bipartite, (b) a subgraph of the join*
 353 *$K_2 \vee mK_1$ for some $m \geq 1$, or (c) K_4 .*

354 **Proof.** Clearly a graph with order at most 4 has an F -WORM coloring using
 355 only 2 colors. We already know that a proper 2-coloring of a bipartite graph is an
 356 F -WORM coloring. If G is a subgraph of $K_2 \vee mK_1$, color the vertices of the K_2
 357 with one color and the other vertices a second color. This gives an F -WORM
 358 coloring.

359 Conversely, let G be a graph with an F -WORM coloring and suppose G is
 360 not bipartite. Then consider any vertex u with at least two neighbors. Since
 361 there is no monochromatic F , it must be that u has a different color to at least
 362 one of its neighbors, say v . Since there is no rainbow F , it follows that every
 363 other vertex has the same color as either u or v .

364 But since G is not bipartite, this means there exists an edge xy where x and y
 365 have the same color, say color 1. By the lack of monochromatic F it follows that
 366 every other vertex has the other color, say 2. If the vertices of color 2 form an
 367 independent set, then we have a subgraph of the join $K_2 \vee mK_1$ for some $m > 0$.
 368 Otherwise, by the same reasoning there are exactly two vertices of color 2 and
 369 we have a subgraph of K_4 . ■

370 6. EXTREMAL QUESTIONS

371 The classical Turán problem ask for the maximum number of edges of a graph
 372 with n vertices that does not contain some given subgraph H . This maximum

is called the Turán number of H . Here we consider an analogue of the classical Turán problem: what is the maximum number of edges of a graph with n vertices if the graph admits a F -WORM coloring? We will let $wex(n, F)$ denote this maximum. We have the following result when $F = P_3$.

Theorem 20. *For $n \geq 1$,*

$$wex(n, P_3) = \begin{cases} \frac{n(n+2)}{4} & \text{if } n \text{ is a multiple of } 4 \\ \frac{n^2+2n-4}{4} & \text{if } n \equiv 2 \pmod{4} \\ \frac{(n-1)(n+3)}{4} & \text{otherwise.} \end{cases}$$

Proof. Consider a graph G that has a P_3 -WORM coloring. By Theorem 4, there is such a coloring using only two colors, say red and blue. Such a coloring is a P_3 -WORM coloring if and only if there is no monochromatic P_3 . It follows that the maximum number of edges in G is obtained by taking some complete bipartite graph and adding a maximum matching within each partite set.

When n is a multiple of 4, then the number of edges is obviously maximized when the two colors are used equally. When n is odd, the maximum is when the two colors are used as equally as possible. When n is even but not a multiple of 4, there are actually two extremal graphs: adding $n/2$ edges to $K_{n/2-1, n/2+1}$ or adding $2(n/2 - 1)/2$ edges to $K_{n/2, n/2}$. We omit the calculations. ■

7. CONCLUSION

We have considered WORM colorings where the forbidden graph is P_3 and provided existence results and bounds on the maximum number of colors for several graph families. The next step would be to consider other forbidden graphs, such as K_3 . Indeed a natural generalization is to consider sets of forbidden graphs, such as all cycles. Some results in this direction are given in [6].

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